

CO3- ASSESSMENT TOOL 2

SIMULATION TASK

COURSE NAME: DIGITAL SIGNAL PROCESSING

COURSE CODE: ECA0910

7. Limit Cycle Oscillation in IIR Filter

Simulate a second-order fixed-point IIR filter and investigate the occurrence of limit cycle oscillations after the input becomes zero. (25) (BL4) (WK4)

Given Data:

- IIR Filter: $H(z) = 1/(1 - 1.4z^{-1} + 0.49z^{-2})$
- Word Length = 8 bits
- Input = Zero after 100 samples

Tasks:

1. Implement the filter in fixed-point arithmetic.
2. Apply a finite-duration input.
3. Remove the input after 100 samples.
4. Observe the output response.
5. Identify limit cycle oscillations.
6. Explain the cause of oscillations.



